HCI Notes: Design Considerations, Evaluation, and Human Cognition

HCI - Weeks 5 & 6, Fall 2025

Abstract

These notes provide a comprehensive overview of key design considerations in Human-Computer Interaction (HCI). Part I outlines the fundamental principles of the design process, emphasizing the importance of a user-centered approach and the iterative nature of design. Part II delves into practical screen design and layout, covering principles of grouping, order, alignment, and the use of white space, as well as various navigation structures. Part III focuses on evaluation methods, from traditional metrics to modern qualitative approaches, and presents foundational design heuristics. Finally, Part IV explores the psychological underpinnings of HCI, with a detailed look at human memory systems, models of reasoning and problem-solving, and the role of emotion in interaction.

Contents

Part I

The Philosophy and Process of Design

1 What is Design?

1.1 Achieving Goals Within Constraints (Slide 4)

At its core, design is the process of **achieving goals within constraints**. This involves a careful balancing act between what is desired and what is possible.

- **Goals (Purpose):** This is the "why" of the design. We must ask: Who is this for? Why do they want or need it? The goal defines the intended purpose and value of the product.
- **Constraints:** These are the limitations and forces that shape the design. They can be related to materials (e.g., screen size, available sensors), platforms (e.g., iOS vs. Android vs. Web), time, budget, or technical feasibility.
- **Trade-offs:** Because goals and constraints often conflict, design is a constant process of making trade-offs. For example, adding more features (a goal) might make the interface more cluttered and harder to use (violating a usability goal) and increase development time (a constraint).

1.2 The Golden Rule of HCI Design (Slides 5-6)

The golden rule of any craft is to **understand your materials**. In HCI, the primary materials are computers and people.

1. **Understand Computers:** This means knowing their limitations, capacities, and the tools and platforms available. A designer must understand what is technically possible.
2. **Understand People:** This is the more complex part. It requires understanding:
   - **Psychological aspects:** How people perceive, learn, remember, and solve problems.
   - **Social aspects:** How people communicate and interact in groups.
   - **Human error:** Recognizing that errors are a normal part of human behavior and must be designed for.
3. **Understand their Interaction:** The ultimate focus of HCI is the space where humans and computers meet.

1.3 To Err is Human: A Design Philosophy (Slide 13)

When a physical structure like a concrete lintel breaks under too much weight, we call it a **design error**, not a "lintel error." We know how concrete behaves under stress, and we engineer for it. HCI must adopt the same philosophy for its primary material: the user.

- Accident reports often blame "human error," but this is a symptom, not a cause.
- **Human 'error' is normal and predictable**, especially under stress or cognitive load.
- We know how users behave. Therefore, we must **design for it**. This means creating systems that anticipate common errors and are resilient to them.
- **Central Message:** Treat the user with the same respect and scientific understanding as physical materials. The user is the focus.

2 The Iterative Design Process (Slides 16-18)

Interaction design is not a linear, one-shot process. It is a cyclical and iterative journey of refinement.

Core Principle

You never get it right the first time!

The process involves several key stages in a continuous loop:

1. **Requirements Gathering:** Understanding what is wanted versus what is currently there. This involves user research methods like interviews and ethnography.
2. **Analysis:** Ordering and understanding the requirements, often using tools like task analysis and scenarios.
3. **Design:** Creating a potential solution based on the analysis, guided by principles and guidelines.
4. **Prototyping & Iteration:** Building a preliminary version (prototype) of the design to test ideas. This is the heart of the process, focused on "getting it right" by finding out what is really needed.
5. **Implementation and Deployment:** The final build and release of the product, including documentation and help.
6. **Evaluation:** Assessing the prototype or final product against the requirements. This can be done through heuristic evaluation or user testing, and the findings feed back into the analysis and redesign stages.

2.1 Pitfalls of Prototyping (Slide 94)

Iteration involves "hill climbing"—making small improvements to get to a better design. However, this has risks:

- **Local Maxima:** You can incrementally improve a bad initial idea, but you may only reach the peak of a small hill (the "Malverns") instead of the true optimal solution (the "Matterhorn").
- **To succeed, you need two things:**
  1. A good starting point (based on solid initial research).
  2. To understand *what* is wrong with your current design in order to fix it.

3 Focusing on the User (Slides 20-24)

3.1 Know Your User

The most important rule of user-centered design is to know your users.

- **Who are they?** What are their skills, context, goals, and motivations?
- **They are probably NOT like you!** Designers are often expert users and are deeply familiar with their own product. This makes them a poor proxy for a real user.
- **How to know them?** Talk to them (interviews), watch them (observation), and use your imagination (empathy).

3.2 Personas

A persona is a powerful tool for maintaining user focus throughout the design process.

- **Definition:** A rich, fictional description of an "example" user, synthesized from real user research. It is not a real person but an archetype.

- **Purpose:** It serves as a surrogate user during design discussions, helping to ground decisions in a human context. Instead of asking "What do we think?", the team can ask, "What would Abdul Shakoor think?"
- **Details Matter:** Including specific details—name, age, background, goals, frustrations, and even a photo—makes the persona feel "real" and memorable to the design team.

3.3 Cultural Probes

Sometimes, direct observation is difficult (e.g., in a private home) or intrusive (e.g., with sensitive populations). Cultural probes are a way to gather inspirational data.

- **Probe Packs:** A collection of items (e.g., a disposable camera, postcards with questions, a voice recorder) is given to participants.
- **Process:** Participants use the pack in their own environment over a period of time to record what is meaningful to them.
- **Goal:** The goal is not to gather objective data, but to get impressionistic and inspirational responses that can inform interviews, prompt design ideas, and help designers become "encultured" to the users' world.

4 Scenarios: Stories for Design (Slides 25-33)

Scenarios are rich, narrative stories about users carrying out activities. They are a core tool used throughout the design process.

- **Purpose:**
  - **Communicate:** Stories are a natural and effective way to communicate design ideas to team members, clients, and users.
  - **Validate Models:** You can "play" a scenario against other models (like a task analysis) to see if they hold up.
  - **Understand Dynamics:** They help explore the step-by-step behavior and flow of an interaction.
- **Linearity:** Scenarios are linear, just like time and our lives. This makes them easy to understand. However, a single scenario doesn't show alternative paths or error conditions. Therefore, it is important to use **multiple scenarios** to explore different possibilities.

From a set of scenarios, designers can extract concrete **requirements** for the system.

Part II
Screen Design and Navigation

5 Screen Design and Layout Principles (Slides 60-79)

Good screen layout is not about art; it is about applying basic psychological principles to make information clear, understandable, and easy to use. The fundamental principle is: **form follows function**.

5.1 Available Tools for Layout

Designers have a core set of tools to create structure and clarity:

1. **Grouping of items**
2. **Order of items**
3. **Decoration** (fonts, boxes, colors)

4. **Alignment of items**
5. **White space** between items

5.2 Grouping and Structure

Core Principle of Grouping

If things are logically together, they should be physically together.

This is a direct application of the Gestalt principle of proximity. Use boxes and white space to create clear visual groupings for related information (e.g., separating "Billing details" from "Delivery details").

5.3 Order

The order of items on the screen should match a natural and logical order for the task. For data entry, this is typically top-to-bottom and left-to-right (in Western cultures). The tab order for keyboard navigation must also follow this logical sequence.

5.4 Alignment

Alignment creates clean, crisp lines that make a design easier to scan.

- **Text:** For left-to-right languages, text should be left-aligned. It is readable and easy to scan. Centered or right-aligned text is much harder to read in blocks.
- **Names:** If users are scanning for surnames, the surnames should be aligned. This can be achieved by formatting as 'Lastname, Firstname' or by using two columns.
- **Numbers:** The alignment depends on the purpose. To quickly compare magnitudes, numbers should be right-aligned (for integers) or decimal-aligned (for real numbers). Left-aligning numbers makes comparison very difficult.

5.5 Use of White Space

White space (or negative space) is one of the most important tools in design. It is not "empty" space; it is an active element that can be used to:

- **Separate:** Create clear distinctions between groups of items.
- **Structure:** Form a grid and guide the eye through the page.
- **Highlight:** Surround an important element with white space to draw attention to it.

6 Navigation Design (Slides 35-57)

Navigation design is about helping users find their way around a system. It can be considered at multiple levels:

- **Local Structure:** Navigation within a single screen or from one screen to the next.
- **Global Structure:** The overall information architecture and movement between all screens in the application.
- **Wider Structure:** How the application navigates to and from other applications and the wider environment (e.g., the web).

6.1 Four Golden Rules of Navigation

A good navigation system ensures that at any point, the user knows:

1. **Knowing where you are.** (e.g., breadcrumbs, highlighted menu items)
2. **Knowing what you can do.** (e.g., clear buttons, links)
3. **Knowing where you are going.** (e.g., descriptive link labels)
4. **Knowing where you've been.** (e.g., visited link colors, history)

6.2 Navigation Infrastructures

There are several common patterns for structuring navigation, especially in mobile and web contexts.

- **Hierarchical:** A classic tree structure, where users navigate down from a broad homepage to more specific pages.
- **Nested Doll:** A linear menu pattern common in mobile apps. Users tap through a series of lists to drill down into content. It's good for small screens but makes lateral movement between sections difficult.
- **Hub and Spoke:** A central "hub" screen acts as a launchpad to different, siloed sections ("spokes"). To move between spokes, the user must return to the hub. This is excellent for task-based apps that benefit from focus.
- **Bento Box / Dashboard:** A single, multi-purpose screen aggregates dynamic components and information from various sources. It's a dashboard style where users can unfold layers of additional information.
- **Filtered View:** Unlike a dashboard, this pattern deals with a single dataset. It allows users to explore that data from multiple perspectives using various views, sorts, and filters controlled by the user.

6.3 Modern Navigation Modalities

Interaction has moved beyond the simple point-and-click.

- **Touch & Gesture-Based:** Multi-touch (pinch, swipe), edge gestures, and air gestures (e.g., Kinect).
- **Voice-Based:** Voice commands (Siri, Alexa) and conversational navigation.
- **Sensor & Motion-Based:** Using gyroscopes for tilt/rotation or accelerometers for shake-to-action.
- **Eye & Gaze-Based:** Using eye-tracking to navigate, especially in AR/VR and accessibility tech.
- **Spatial & Immersive (XR):** Using spatial gestures, teleportation, or hand-tracking to navigate in 3D environments (AR/VR/MR).
- **Brain-Computer Interfaces (BCI):** Using EEG to detect brain signals for neural navigation.

Part III
Human Cognition: Memory and Thinking

7 Models of Human Memory

Understanding how human memory works is crucial for designing interfaces that minimize cognitive load.

7.1 The Three-Stage Model of Memory

Memory is often modeled as having three distinct functions:

1. **Sensory Memories:** Brief, transient buffers for stimuli received through the senses (iconic for visual, echoic for aural). They are continuously overwritten. The "sparkler trail" effect is an example of iconic memory.
2. **Short-Term Memory (STM) / Working Memory:** A temporary "scratch-pad" for information we are currently attending to.
   - **Attention** is the gateway from sensory memory to STM.
   - It has rapid access but also rapid decay.
   - It has a famous limited capacity of **7 ± 2 chunks** of information. A chunk is a meaningful unit (e.g., '0121' can be one chunk, not four digits).
3. **Long-Term Memory (LTM):** The vast, repository for all our knowledge.
   - **Rehearsal** is the process of moving information from STM to LTM.
   - It has slower access but a huge, potentially unlimited capacity and very slow decay.

7.2 Types of Long-Term Memory

LTM itself has different forms:

- **Episodic Memory:** Serial memory of events and experiences (e.g., what you did last birthday).
- **Semantic Memory:** Structured memory of facts, concepts, and skills (e.g., knowing that Paris is the capital of France).

Semantic memory is often modeled as a **semantic network**, where concepts are nodes and relationships are links. This structure supports inference (e.g., if a beagle is a type of dog, and a dog has four legs, we can infer that a beagle has four legs).

7.3 LTM Forgetting and Retrieval

- **Forgetting:** Information can be lost from LTM through gradual decay or through interference, where new information replaces old (*retroactive interference*) or old information interferes with new (*proactive inhibition*).
- **Retrieval:** As discussed previously, retrieval from LTM can happen via **recall** (reproducing information from memory, aided by cues) or **recognition** (knowing you've seen something before). Recognition is far less complex and more reliable.

8 Thinking: Reasoning and Problem Solving

8.1 Types of Reasoning

1. **Deductive Reasoning:** Deriving a logically necessary conclusion from given premises. If the premises are true, the conclusion *must* be true. However, a conclusion can be logically valid but factually untrue if the premise is false (e.g., "If it is raining, the ground is dry. It is raining. Therefore, the ground is dry.").
2. **Inductive Reasoning:** Generalizing from specific cases to unseen cases. It is a powerful way we learn about the world, but it is unreliable because it can only be proven false, not true. (e.g., "All swans I have seen are white, therefore all swans are white.").
3. **Abductive Reasoning:** Reasoning from an event to a likely cause. It is a form of inference to the best explanation, but it is also unreliable and can lead to false conclusions. (e.g., "Sam drives fast when drunk. I see Sam driving fast, so I assume he is drunk.").

8.2 Problem Solving

Problem solving is the process of finding a solution to an unfamiliar task.

- **Problem Space Theory:** Describes problem solving as a process of generating states using legal operators to move from a start state to a goal state, guided by heuristics.
- **Analogy:** Using knowledge from a similar problem in a familiar domain to solve a novel problem.
- **Skill Acquisition:** Experts solve problems differently from novices. They chunk information more effectively and group problems based on deep conceptual structure, not superficial features.

8.3 Errors: Slips vs. Mistakes

- **Slips:** A user has the right intention but fails to perform the action correctly due to inattention or poor physical skill.
- **Mistakes:** A user has the wrong intention because their understanding of the system (their mental model) is incorrect.

9 Emotion

Emotion and cognition are deeply intertwined. Affect (the biological response to stimuli) influences how we think.

- **Positive Affect:** Encourages creative problem solving.
- **Negative Affect (Stress):** Narrows thinking and increases the difficulty of even simple tasks.

Implication for Design: Aesthetically pleasing and rewarding interfaces can increase positive affect, making users more relaxed, forgiving of minor flaws, and better at solving complex problems.